Matty Aldridge

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Education

University of Edinburgh, MEng Mechanical Engineering 2025 – 2029
Direct entry into second year; now in third year. Placement slot January to August 2028.

- Modules include fluid mechanics, structural mechanics, materials science, manufacturing processes, mechatronics and engineering mathematics.

Nanjing University of Aeronautics and Astronautics, student engineering delegate Jun – Jul 2026

- Four-week aerospace programme: flight dynamics, advanced manufacturing, lightweight materials and drones.

Hills Road Sixth Form College, Cambridge 2023 – 2025

- A-levels: Mathematics, Further Mathematics, Physics, Chemistry (A*A*A*A); Extended Project A*. Ten GCSEs at grade 9 (Impington Village College).

Research and Engineering Experience

Thermal-mechanical engineer, Project Xenith, Endeavour Oct 2026 – present
University of Edinburgh rocketry society

- Working on the thermal and mechanical design of the team's Hall-effect thruster. Built its fully parametric SolidWorks assembly: 32 native parts driven by 99 global variables, so the whole thruster updates as the team's design values change.

Powertrain engineer, Edinburgh University Formula Student Sep 2025 – present

- One of a seven-person powertrain team in 2025/26, the year EUFS became the first British team to fully pass autonomous (ADS) scrutineering.
- Owned the SolidWorks design of the car's low-voltage electronics box and its PCB layout in Altium; added fillet radii at sharp internal corners to remove a stress concentration.

Research intern, ML systems (MLIR / C++) Jul – Oct 2026
School of Informatics, University of Edinburgh. Supervisor: Dr Ajitha Rajan

- Built an out-of-tree MLIR metamorphic testing and verification suite for an 8-bit quantisation compiler: custom transform passes across the linalg, affine, memref and arith dialects, a bit-exact JIT oracle over the MLIR-to-LLVM lowering, and AFL++ fuzzing.
- Extended the approach to Google's HEIR homomorphic-encryption compiler, fuzzing the CKKS and BGV pipelines with real OpenFHE encryption. Reported compiler bugs upstream (google/heir #3384, #3386) and submitted a fix as a pull request (#3385).

Projects

Open-source aerospace and thermal solver suite (Python) 2026

- 18 first-principles solvers with a public web portal, covering re-entry heating, scramjet and aerospike cooling, a SABRE-style precooler, tokamak divertor heat loads, spacecraft thermal and attitude control, and truss FEM.
- Oblique shock and Prandtl–Meyer solver matches NACA Report 1135; normal-shock relations reproduced exactly. Built a metamorphic test harness that checks 8 physical invariants across three of the solvers.

LEO satellite tracking ground station Apr 2026 – present

- Designing and building a dual-axis (azimuth/elevation) tracking mount for a parabolic grid dish: 57:1 planetary drive with 1/256 microstepping for sub-millidegree step resolution, IMU heading feedback, slip ring and full parametric CAD.

Awards

- Winner, Royal Aeronautical Society Marshall Lecture Junior Competition (2024).
- Merit, British Mathematical Olympiad and British Physics Olympiad (2024).

Skills and Other

Software	Python (NumPy, SciPy), C++, MLIR/LLVM, PyTorch, Git, Docker, AFL++
Engineering	SolidWorks, Altium Designer, Ansys Mechanical
Languages	English (native), French (B1), Czech (conversational)
Other	Founder of pringlesmaths, a maths account with 13,000 followers; INCiTE summer school delegate, Yogyakarta (2026)